

Sucharita Chattopadhyay

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Education

VIT-AP University

B.Tech in Computer Science Engineering

- **Current CGPA: 9.79/10** • CSE Branch Rank Holder
- Solved **200+ Data Structures & Algorithms problems in Java** across LeetCode and GeeksforGeeks — arrays, hashing, two-pointer, and sliding-window patterns.

Amaravati, Andhra Pradesh

Aug 2024 – Jul 2028

D.A.V. Public School — Higher Secondary (PCMB): 92.2%

Delhi Public School — Secondary (CBSE Class X): 98.8%

Bankura, West Bengal

Rourkela, Odisha

Technical Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C/C++

CS Foundations: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks (TCP/IP), SDLC, Agile/Scrum

Reliability, Testing & Observability: pytest, unit & integration testing, code reviews, test harness design & coverage analysis, monitoring & observability, performance optimization, root-cause analysis, systematic debugging, fault-tolerant pipelines, rate-limit & failure handling

API & Systems Design: REST API design, FastAPI, HTTP request lifecycle, webhooks & event-driven processing, caching, database schema design & migrations, pgvector semantic search

Secure Development: secure coding best practices, JWT authentication & authorization, HTTPS/TLS, input validation (Pydantic), secrets management

Infrastructure & Delivery: Docker, Linux, Google Cloud Platform (GCP), Git, CI/CD, high-availability & zero-downtime deployment

Frontend & Applied AI: React, Next.js, Tailwind CSS; RAG, multi-agent orchestration & tool calling (LangGraph, LlamaIndex), OpenAI & Gemini (Vertex AI) APIs

Experience

Open Paws

AI Engineer Intern

Bengaluru, India

May 2026 – Jul 2026

- Resolved 35+ tracked bugs across SQL migrations, timezone logic, and caching, and built a 55-test pytest harness (6 suites) reaching 100% coverage on the email delivery and rendering services.
- Cut manual legislative research effort by 35% by designing concurrent multi-agent LLM workflows (LangGraph, LlamaIndex, OpenAI APIs) with RAG and tool calling — working with the non-technical advocacy client to determine user requirements and brief stakeholders on design trade-offs.
- Shipped a Python data-ingestion service syncing 300+ legislative bills daily from the congress.gov API, with historical backfill, rate-limit handling, and PDF-fallback parsing that kept ingestion reliable and available through malformed upstream data.
- Delivered 10+ production features across a 46-endpoint FastAPI backend and Next.js UI on GCP — pgvector semantic search, dual relevance scoring, personalized email digests, and secure JWT-authenticated access — collaborating in Agile sprints and code reviews on a distributed team.

AARC — Code for Compassion Program

Web Developer (Selected Program Participant)

Bengaluru, India

Feb 2026 – Mar 2026

- Engineered VERDE, a sustainability platform processing 150+ menu combinations into per-menu sustainability impact comparisons via Node.js REST APIs, enabling cafes to simulate plant-based menu alternatives.
- Automated releases via Git-based CI/CD on Vercel serverless (cloud) infrastructure, enabling atomic zero-downtime deploys of the TypeScript/Next.js/React/Tailwind frontend.

Projects

PayFlow — Multi-Gateway Payment Orchestrator (System Design + Frontend)

github.com/Sucharita2006/PayFlow

- Architected a payment-orchestration system integrating 3 payment gateways (Stripe, Razorpay, PayU) with a priority-based 3-level fallback chain, Redis config cache, and event-driven webhook intake — designed to maximize transaction success and availability while minimizing processing fees.
- Implemented the operations & observability dashboard (React, TypeScript, Tailwind CSS) surfacing 8+ real-time monitoring metrics across 5 visualization components — traffic, success/failure rates, latency, and gateway health — plus transaction search and filtering across gateways and statuses.

LegiTrack AI — Animal Legislation Intelligence Platform

github.com/Sucharita2006/LegiTrack-AI | legi-track-ai.vercel.app

- Independently built a full-stack legislative-intelligence platform (FastAPI, SQLAlchemy, React/TypeScript) operating at scale across all 50 US state legislatures, processing up to 1,000 bills per run through an LLM stance classifier and a deterministic 100-point urgency-scoring algorithm.
- Improved pipeline efficiency and performance, cutting LLM API costs by 60–80% with an NLTK keyword-gatekeeping pre-filter and async GraphQL ingestion, ensuring only relevant bills trigger costly model calls.
- Validated LLM classification at 93.75% accuracy via pytest unit tests (scoring, ingestion) and manual audits across 50-state data, persisting normalized records to a SQLite/SQLAlchemy store behind filtered, paginated REST endpoints.

Leadership & Achievements

- **District Topper** — Sundergarh District, Class X Board Examinations, with 100/100 in five subjects.
- **Post-a-thon Hackathon Finalist** — Led a team to Top 5 among 90+ teams with a sustainability-focused web solution.